

Tien Dung Vu

AI ENGINEER / DATA SCIENTIST

+61 466 599 354

tiendung7v1@gmail.com

linkedin.com/in/tiendungvu

billvu05.github.io

Epping, NSW, Australia

PROFESSIONAL SUMMARY

Builds and ships production AI systems: LLM agent pipelines, RAG assistants, and ML models deployed to AWS behind FastAPI endpoints. Works the analytics side too: ETL into PostgreSQL, hypothesis testing and A/B tests reported with confidence intervals. Builds the evaluation harness before trusting the model, so quality is a number per change rather than an opinion.

WORK EXPERIENCE

AI Automation Intern

May 2026 - Present

Affinity Marketing Australia - Sydney, Australia

- Implement AI in live business workflows, connecting **Claude** to **n8n**, HubSpot and campaign channels so LLM-powered automations replace manual steps in day to day CRM, content and advertising operations
- Shipped **4 end to end LLM content pipelines** in n8n, calling the Claude API from webhook triggered workflows to eliminate manual production steps across recurring campaign cycles
- Engineered an email subject line **A/B testing** pipeline and a weekly analytics summary emailer, automating recurring campaign testing and reporting workflows
- Built a CRM data entry automation with Claude Code, formatting practitioner, client and contract information into HubSpot across **500+ contact records** and cutting data formatting error rates by **50%**

AI Engineer Intern

Oct 2024 - Apr 2025

CMC Corporation - Hanoi, Vietnam

- Engineered the AI layer of an employee onboarding system: a **retrieval augmented generation (RAG)** assistant over **700 internal policy documents (~3,000 pages)**, embedding the corpus and serving semantic retrieval from a **FAISS vector index** into LLM APIs with prompt engineering, letting **50 HR staff** query procedures in natural language
- Deployed trained models to production on AWS** behind **10 FastAPI endpoints**, containerised with Docker, so the system consumed predictions through a stable cloud API rather than ad hoc scripts
- Built classical ML models with scikit-learn and **XGBoost** for propensity and attrition risk scoring, K-Means clustering to segment employee cohorts, and **time series models** forecasting onboarding volume and resourcing demand
- Built ETL/ELT pipelines in Python and SQL loading **1,200+ HR, onboarding and operational records** into PostgreSQL on daily incremental scheduled loads, with schema validation and deduplication
- Ran **hypothesis testing and A/B tests** on onboarding interventions, reporting lift with confidence intervals so decisions rested on measured effect rather than anecdote

PROJECTS

LLM Research Agent (CLI)

Python, LangGraph, Google Gemini, NumPy, Docker, Pytest, GitHub Actions | Jul 2025

- Built a CLI research agent turning a natural language question into a cited answer as strict JSON, implemented as a LangGraph StateGraph over six nodes with a typed state schema and a conditional edge that routes back to search when retrieved context fails to cover a required fact, covered by an **87-test Pytest suite** and GitHub Actions CI
- Built an **evaluation harness** over a 25-question golden set scoring citation validity, keyword recall, slot fill rate, LLM calls per question and p95 latency, with a context recall metric separating a search that never found a fact from a model handed it and dropping it
- Eliminated a class of citation hallucination by reconstructing citations from retrieved context using the source indices the model returns, so a cited URL is provably one the agent fetched

Pacific Wings Airline Strategy Simulator

Python, FastAPI, XGBoost, scikit-learn, LangGraph, PostgreSQL, Docker | Jun 2026 - Present

- Built a **model selection pipeline for demand forecasting** scoring a seasonal index model and XGBoost against two naive baselines on a forward looking 2025 holdout, deploying the winner only if it beats the best baseline by a set margin: the seasonal index model reached **MAPE 4.37% (R²=0.997)**, ahead of XGBoost (9.27%) and the same month last year baseline (4.70%)
- Replaced a leaky shuffled k-fold with **leave one route out cross validation**, built prediction intervals from rolling origin out of fold residuals, and designed a **confidence scoring system** combining 200-resample bootstrap spread, per route forecast reliability and extrapolation distance into one explainable 0-100 score
- Implemented a **LangGraph five agent pipeline** where deterministic Demand and Finance agents extract figures from the simulation engine and Gemini agents narrate them, so reported numbers stay real even when commentary degrades

Australian Media Sentiment and Framing of AI

Python, VADER, RoBERTa, FinBERT, BERTopic, Guardian API | Jan 2026 - Jun 2026

- Built an ETL pipeline collecting **2,622 Guardian Australia articles** via the Guardian Open Platform API with rate limited pagination and multi-criteria relevance filtering, reducing the corpus to **1,033 articles** spanning 2017 to 2026
- Applied a three model sentiment framework (VADER, RoBERTa, FinBERT) plus a FinBERT sliding window over full article bodies, detecting a persistent headline to body sentiment divergence across all 10 major Guardian sections
- Quantified inter model agreement across 1,033 articles and applied **BERTopic** to identify dominant risk themes, delivering **6 evidence based editorial recommendations** to The Sydney Morning Herald

EDUCATION

Bachelor of Artificial Intelligence, University of Technology Sydney (UTS)

Graded: High Distinction

SKILLS

Programming: Python, SQL

AI & Machine Learning: PyTorch, scikit-learn, XGBoost, NLP, Named Entity Recognition, Time Series Forecasting, Clustering (K-Means), Text Classification, Sentiment Analysis

LLM & Agentic Systems: LangGraph, RAG, Embeddings, Vector Search, FAISS, Semantic Retrieval, Prompt Engineering, Agentic Workflow Design, LLM Pipeline Design

Data & Databases: ETL/ELT Pipelines, PostgreSQL, Data Quality Assurance, Feature Engineering

Backend & APIs: FastAPI, REST APIs, System Design, Object Oriented Programming

Cloud & Deployment: AWS, Docker, Model Serving, Containerised Deployment, CI/CD, GitHub Actions

Analytics & Experimentation: Hypothesis Testing, A/B Testing, Confidence Intervals, Cross-Validation, Model Evaluation Metrics (AUC-ROC, Precision/Recall/F1)

BI & Reporting: Power BI, Looker Studio, GA4, Advanced Excel (Power Query, XLOOKUP, Solver)